

Mock Exam — Weeks 1 to 3

DSA27 · Data Structures and Algorithms · 60 marks: Section A 30 · Section B 30

Question bank practice paper

Fall 2026

How to use this paper. Sit it in one go, closed book, with no computer. The questions are taken from the weekly question banks, so the model answers are in [week01-answers.md](#), [week02-answers.md](#) and [week03-answers.md](#) — look them up **only after** you have finished and marked yourself. Its format follows the real final: **30 marks of multiple choice, 30 marks of written answers**. It covers only weeks 1–3; the real exam covers the whole course.

Section A — Multiple choice (30 marks)

One mark each. Choose one answer out of four. Write the letters on your answer sheet.

#	Question	#	Question	#	Question
1	W1-M02	11	W2-M02	21	W3-M02
2	W1-M06	12	W2-M03	22	W3-M04
3	W1-M10	13	W2-M06	23	W3-M06
4	W1-M11	14	W2-M07	24	W3-M08
5	W1-M15	15	W2-M11	25	W3-M09
6	W1-M17	16	W2-M12	26	W3-M12
7	W1-M19	17	W2-M16	27	W3-M15
8	W1-M21	18	W2-M18	28	W3-M16
9	W1-M23	19	W2-M21	29	W3-M17
10	W1-M24	20	W2-M23	30	W3-M22

The questions are in [week01-questions.md](#), [week02-questions.md](#) and [week03-questions.md](#).

Section B — Written answers (30 marks)

Answer **all** questions. Marks are shown in brackets.

#	Question	Topic	Marks
B1	W1-E4	ADT against data structure, with the Stack	4
B2	W2-E4	why array indexing is $O(1)$ and insertion $O(n)$; fixed size	4
B3	W2-P3	prove $4n^2 + n = \Theta(n^2)$	4
B4	W2-K3 and W2-K5	complexity analysis, 2 marks each	4
B5	W2-S2	the forward-shifting insertion bug, traced	3
B6	W3-E2	naive against memoised Fibonacci	4
B7	W3-R3	recurrence, solution and stack depth of <code>range_sum</code>	4
B8	W3-T3	trace <code>gcd(48, 18)</code> and its call stack	3
Total			30

Marking yourself

- **Section A:** one mark per correct letter. The answer tables are at the top of each weekly answers file.
- **Section B:** use the **marking points** in the answers files. Award a mark for each point your answer makes clearly and correctly — not for length.
- **Then** look at *which* questions you lost marks on, and on which level ([what], [how] or [why]). That tells you what to revise, and how.